

Introduction to Computational Thinking and Python Programming

Session 4

Instructor: Sarom Leang, PhD
Specialist: Daniela Garcia Galindo
Innovation Hall

Pronouns

Instructor

Sarom Leang, Ph.D.

Pronouns: Teacher / Instructor

Specialist

Daniela Garcia Galindo

Pronouns: Daniela

Schedule



10:00 AM - 10:15 AM

Homeroom



10:15 AM - 11:35 AM

G1



11:40 AM - 12:35 PM

Lunch



12:40 PM - 02:00 PM

G2

Student Expectations



NO FOOD



**NO DRINKS
(on the table)**



Be respectful to
individuals and
property



Be open to
learning



Be open to not
understanding



Be patient with
yourself



Ask questions



Explore



Embrace failure

Course Overview

This course introduces the fundamental building blocks of computational thinking and computer programming using the Python language.

Upon successful completion of this course, students will be able to:

- Improve their problem-solving skills
- Write, read, and execute Python code using basic data types and operators

Course Overview (cont.)

Exploratory Topics

- ~~Session #1 – November 1, 2025~~
 - ~~Entrance Survey~~
 - ~~How Computers Work~~
- ~~Session #2 – December 6, 2025~~
 - ~~Parallel Computing~~
- ~~Session #3 – January 31, 2025~~
 - ~~Generative Artificial Intelligence~~
- Session #4 – March 7, 2026
 - Generative Artificial Intelligence
- Session #5 – April 11, 2026
 - Future of Computing (Quantum)
 - Exit survey

Topics

- Algorithms and Problem Solving
- Debugging and Troubleshooting
- Loops
- Algorithms and Syntax
- Variables and Conditionals
- Variable Arithmetic
- Conditionals (If/Else)
- Compound Conditionals

NEWS



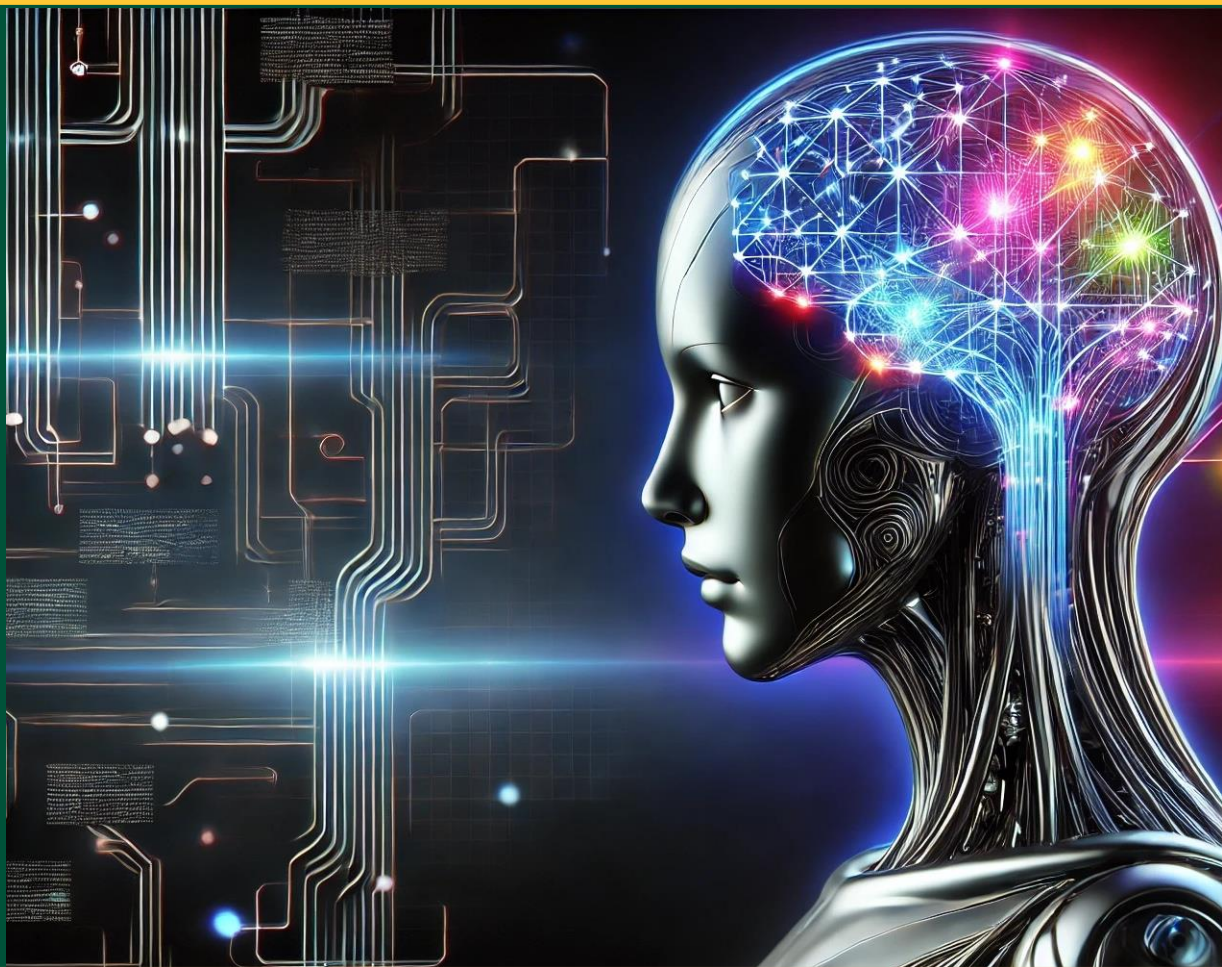
Technology in the News

STEM Fusion

How Large Language Models Work

AI DEEP DIVE

LLM Basics · Tokenization · Training
Predictions · Prompting · AI Ethics



Overview

01



What is AI & LLMs?

02



Tokenization

03



Training Data

04



Predictions & Probability

05

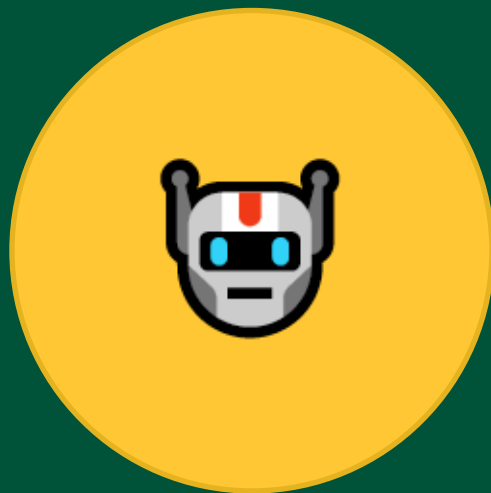


Prompting

06



AI Ethics & Academia



What is AI & LLMs?

From Siri to ChatGPT — what's actually going on 🤖

What Is Artificial Intelligence?

Artificial Intelligence

- Software that learns patterns from data
- Finds connections humans might miss
- Gets better with more examples
- Does NOT "think" — it calculates
- Examples: Siri, Gmail spam filter, Netflix recommendations, TikTok For You

Regular Software vs. AI

Regular software follows exact rules a programmer wrote.

AI learns rules from millions of examples — nobody wrote a rule for every sentence ChatGPT can say.

Think of teaching a child to recognize dogs: you don't list rules, you show pictures. AI works the same way.

AI vs. Machine Learning vs. Deep Learning



Artificial Intelligence (AI)

Any technique that lets machines perform tasks that typically require human intelligence — planning, reasoning, problem-solving, perception, language.



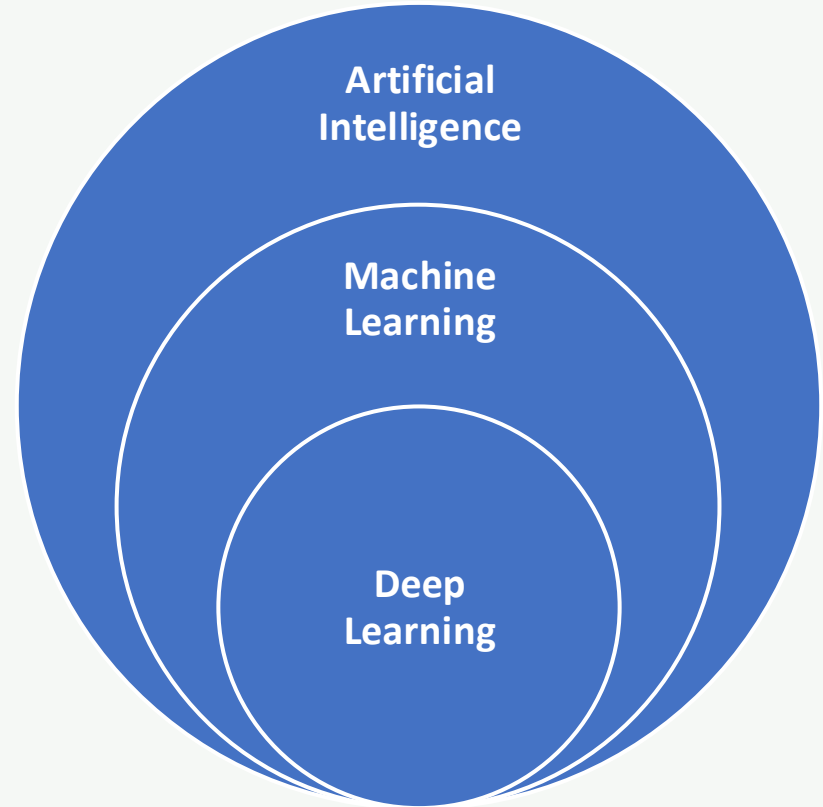
Machine Learning (ML)

A subset of AI where the machine learns from data and improves automatically — no programmer writes explicit rules for every decision.



Deep Learning (DL)

A subset of ML using multi-layer neural networks. Powers image recognition, speech, and LLMs. Needs lots of data and compute.



What Is a Large Language Model (LLM)?

1

Trained on billions of sentences from the internet, books, Wikipedia, and code



2

Learns patterns — what words tend to follow other words



3

Can generate new text that sounds like human writing



4

Powers ChatGPT, Claude, Gemini, Copilot, and more



In short: LLMs are enormous text-prediction machines — trained on the internet at a scale hard to imagine.

What Is Generative AI?

Generative AI — technology that learns statistical patterns in data and uses that knowledge to create new content that resembles its training.

Text



ChatGPT, Claude,
Gemini

Images



DALL-E, Midjourney,
Adobe Firefly

Music



Suno AI, AIVA,
Mubert

Video



Sora, Runway,
Descript

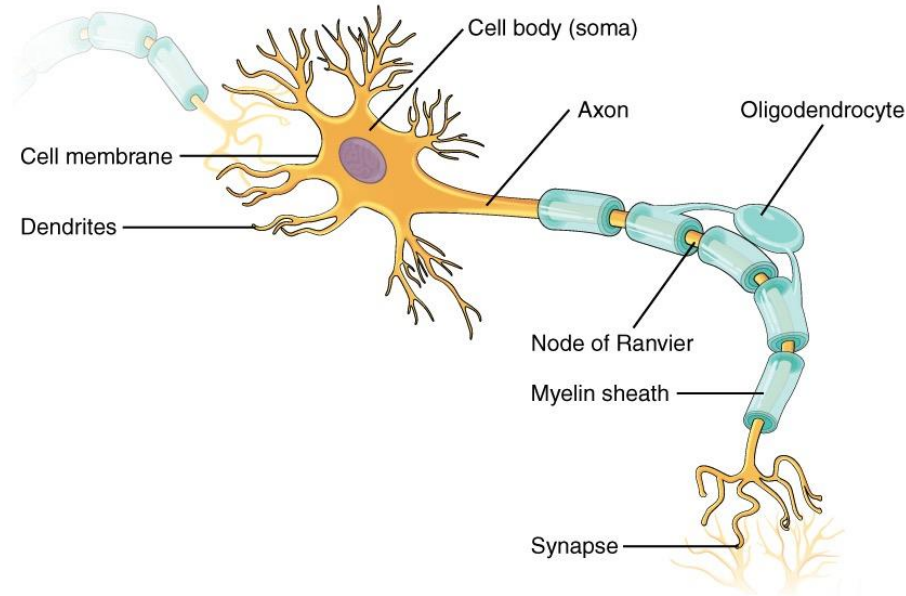
All four work the same way: learn patterns from massive datasets → generate new content that resembles the training data



Neural Networks

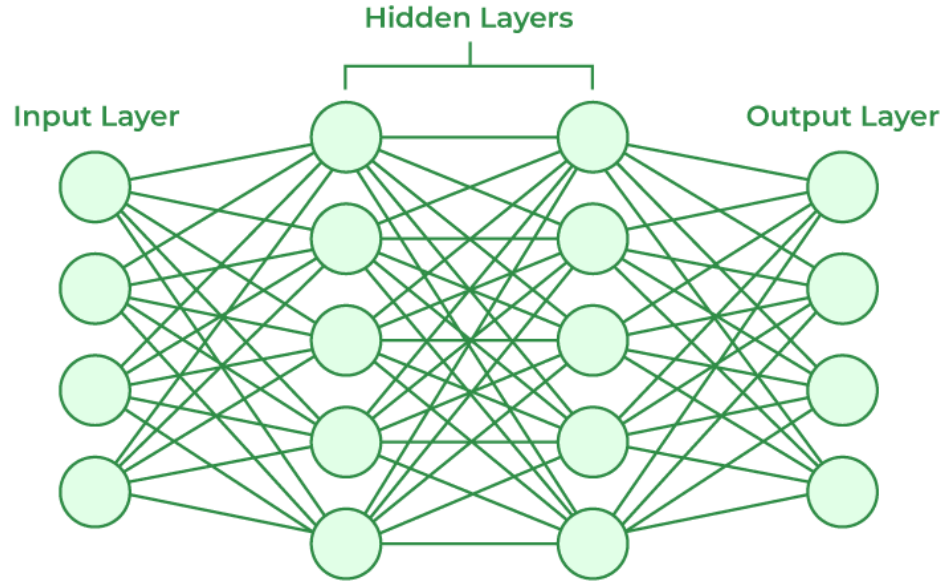
Neural Networks (Neuroscience)

- Networks of nerve cells (**neurons**) in the brain and nervous system that are interconnected and **communicate through synapses** to **process information, control bodily functions, and enable behaviors**



Neural Networks (Computer Science)

- Also referred to as artificial neural networks, loosely modeled on the organization of the human brain, form the basis for many artificial intelligence systems



From Brain Cell to Math: The Neuron Analogy

Biology inspired Computer Science



Biological Neuron

Dendrites

Receive incoming signals from other neurons

Nucleus

The cell body — processes all the incoming signals

Synapses

Connection junctions that control signal strength

Axon

Sends the output signal to the next neuron



Artificial Neuron

Inputs

Numbers representing each token or feature fed into the node

Node

Multiplies each input by its weight and sums everything up

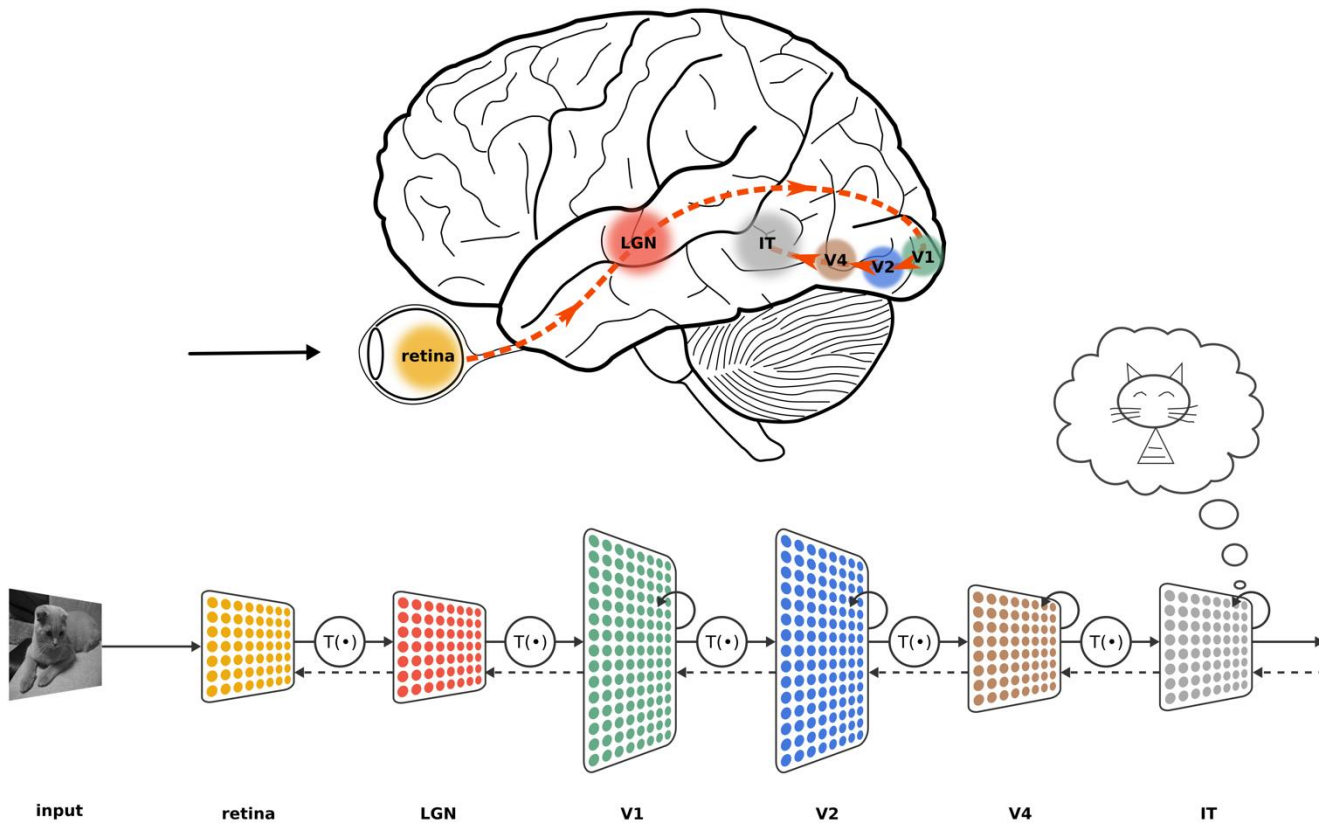
Weights

Numbers that control how much each input matters — learned during training

Output

A single number passed forward to nodes in the next layer

Artificial Neural Networks



Neural Networks: The Engine Inside LLMs

Key Concepts

⚡ Neuron (Node)

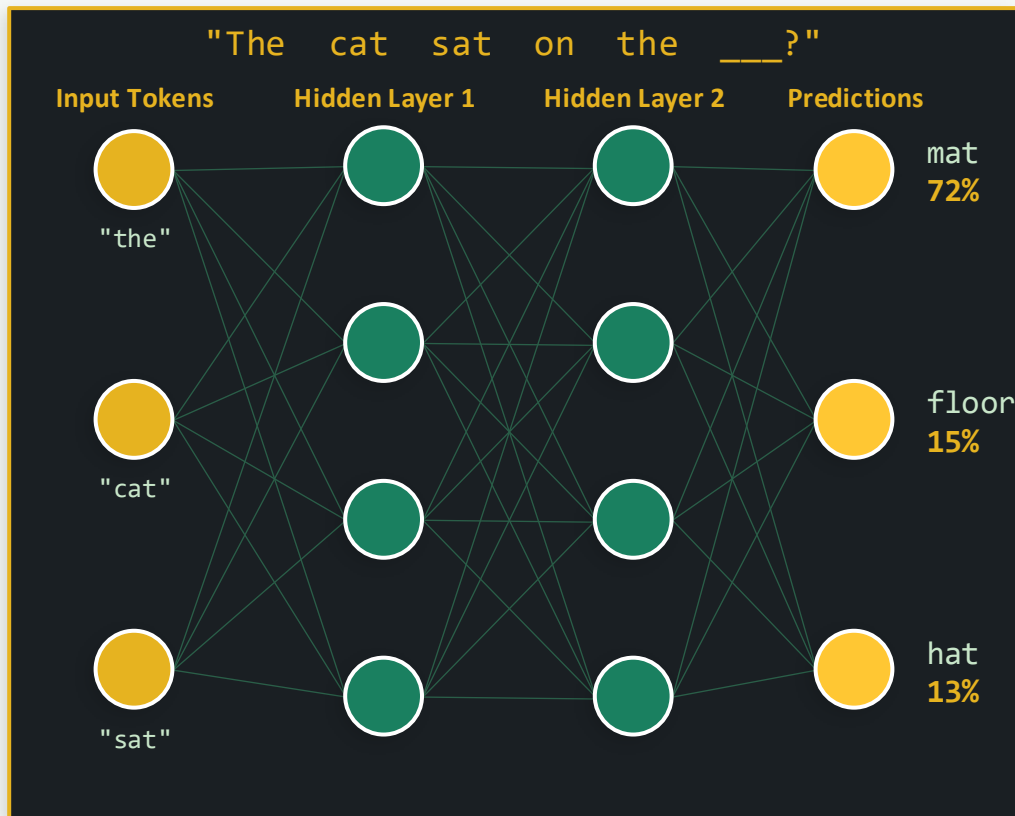
A tiny math unit. It takes several numbers in, multiplies each by a weight, adds them all up, and passes one number forward to the next layer.

⚖ Weight

A number controlling how strongly one neuron influences another. Training fine-tunes billions of weights to minimize prediction errors.

📖 Layer

A group of neurons working in parallel. LLMs stack 96+ layers — early layers detect word shapes; deeper layers learn grammar, facts, and reasoning.



Generative AI

- A type of artificial intelligence that can **create new content that is similar to the original data it was trained on**
- **Learns patterns and relationships in data, and uses that knowledge to generate new content**

Data > Information > Knowledge

Data (raw data)	Information (processed and organized)	Knowledge (insight gained)
Census records	Demographics	What will be the average age of the population in 2040?
Weather measurements	Average temperature, rainfall, etc. in a region over a given time range	How will the weather be next week, month, year?
Financial records	Growth rate of a company	Where will this company be in the next five years?
Patient records	Incident rate of a disease	When will COVID-19 end?

Famous LLMs You Already Know

ChatGPT

Made by: OpenAI

Chatbot, writing, coding help

Claude

Made by: Anthropic

Research, analysis,
long writing

Gemini

Made by: Google

Search, Gmail, Google Docs AI

Copilot

Made by: Microsoft

Word, Excel, coding assistance

Llama

Made by: Meta

Open-source, research use

Siri / Alexa

Made by: Apple / Amazon

Voice assistant, smart home

Generative AI Tools You Can Use Today



Text Generation

ChatGPT (OpenAI)

Claude (Anthropic)

Gemini (Google)

Write, summarize, explain, translate, code, brainstorm — and more.



Image Generation

DALL-E

Midjourney

Stable Diffusion

Adobe Firefly

Type a description → get a unique image. No drawing skills required.



Music Generation

Suno AI

AIVA

Mubert

Beatoven

Describe a mood or genre → full AI-composed track in seconds.



Video Generation

Sora (OpenAI)

Runway

Descript

Capsule

Generate short video clips from text prompts — still early-stage.

Generative AI: How Fast Things Are Moving

	Pre-2020	2020	2022	2023	2025	2030
Text	Spam filter Translation Basic Q&A	Copywriting First drafts	Longer form Second drafts	Fine-tuned for science	Better than human average	Better than pro writers
Code	1-line autocomplete	Multi-line generation	Longer form Better accuracy	More languages	Text-to- product (draft)	Better than full-time devs
Images	—	—	Art Logo Photos	Mock-ups Product design Architecture	Final drafts Product design	Better than pro designers
Video / 3D	—	—	First 3D / video models	Basic draft video & 3D	Second drafts	AI Roblox Personalized films & games

Source: adapted from industry research and AI capability tracking. Timelines are estimates — the field moves faster than most predictions.

⚡ OPTIONAL CHALLENGE — What Is AI?



Pen & Paper

Think about your day so far.

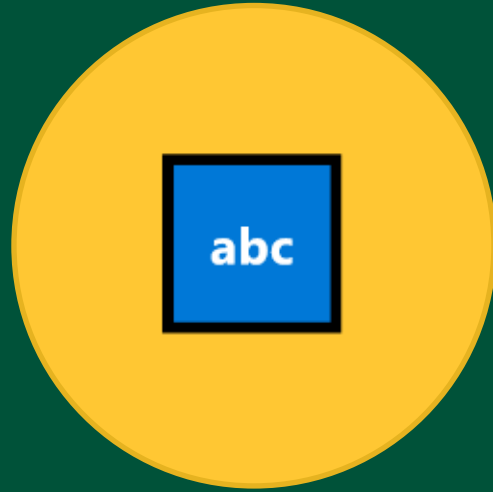
1. List 3 places you interacted with AI
(Hint: phone, apps, school tools).
2. For each one, write:
 - What did the AI DO?
 - What data do you think it learned from?
3. BONUS: Which is an LLM vs. a simpler AI?
 - Google autocomplete
 - Netflix "Recommended for you"
 - ChatGPT answering a question
 - Face ID on your phone

Explain your reasoning for each.



Try it with an AI Tool

```
# Try asking an AI tool directly:  
# "Are you an LLM? How do you work?"  
#  
# Then ask:  
# "What's the difference between  
# you and a Google search?"  
#  
# Write down the most surprising  
# thing it tells you.
```



Tokenization

How AI reads text — one chunk at a time



What Is a Token?

A token is a chunk of text — usually a word, part of a word, or punctuation. AI models don't read whole sentences at once — they convert text into a list of tokens first.

Example: "Hello, world!" → 4 tokens:

"Hello"

Token 1

", "

Token 2

" world"

Token 3

!"

Token 4

 Rule of thumb: ~4 characters = 1 token. "ChatGPT is amazing!" ≈ 6 tokens. A 1,000-word essay ≈ 1,300 tokens.

Why Does Tokenization Matter?

Context Window

AI can only 'see' a limited number of tokens at once. GPT-4 handles ~128,000 tokens (about 90,000 words). Shorter inputs are processed faster.

Cost & Speed

API calls to AI are charged per token. Engineers optimize prompt length to save money. Knowing this helps you understand why AI products have message limits.

Languages Differ

English words are ~1 token each. Some languages (Japanese, Arabic) use more tokens per idea. The same sentence can cost more in some languages.

Numbers Are Tricky

"12345" might split into ["123", "45"] — two tokens! This is one reason AI can struggle with exact arithmetic on large numbers.

Tokenization in Action – Let's Break It Down

Watch how different sentences tokenize:

"I love pizza"



"I"

" love"

" pizza"

3 tokens

"Unbelievable!"



"Un"

"bel"

"iev"

"able"

"!"

5 tokens

"12345"



"123"

"45"

2 tokens

"supercalifragilistic"



"super"

"cal"

"ifrag"

"ilis"

"tic"

5 tokens



Try it live at: platform.openai.com/tokenizer

⚡ OPTIONAL CHALLENGE — Tokenization



Pen & Paper

Predict the token count — then check!

1. Guess how many tokens each has:

- a) "Hello!"
- b) "The quick brown fox"
- c) "supercalifragilistic"
- d) "2024"
- e) "🦄"

2. Which do you think uses MORE tokens:

- A 10-word English sentence, OR
- The same 10-word sentence translated to Japanese?

Why do you think so?



Try it with an AI Tool

```
# Go to: platform.openai.com/tokenizer
#
# Type each sentence and count:
# "Hello!"
# "The quick brown fox"
# "supercalifragilistic"
# "2024"
# An emoji of your choice
#
# Were your guesses correct?
# What surprised you most?
```

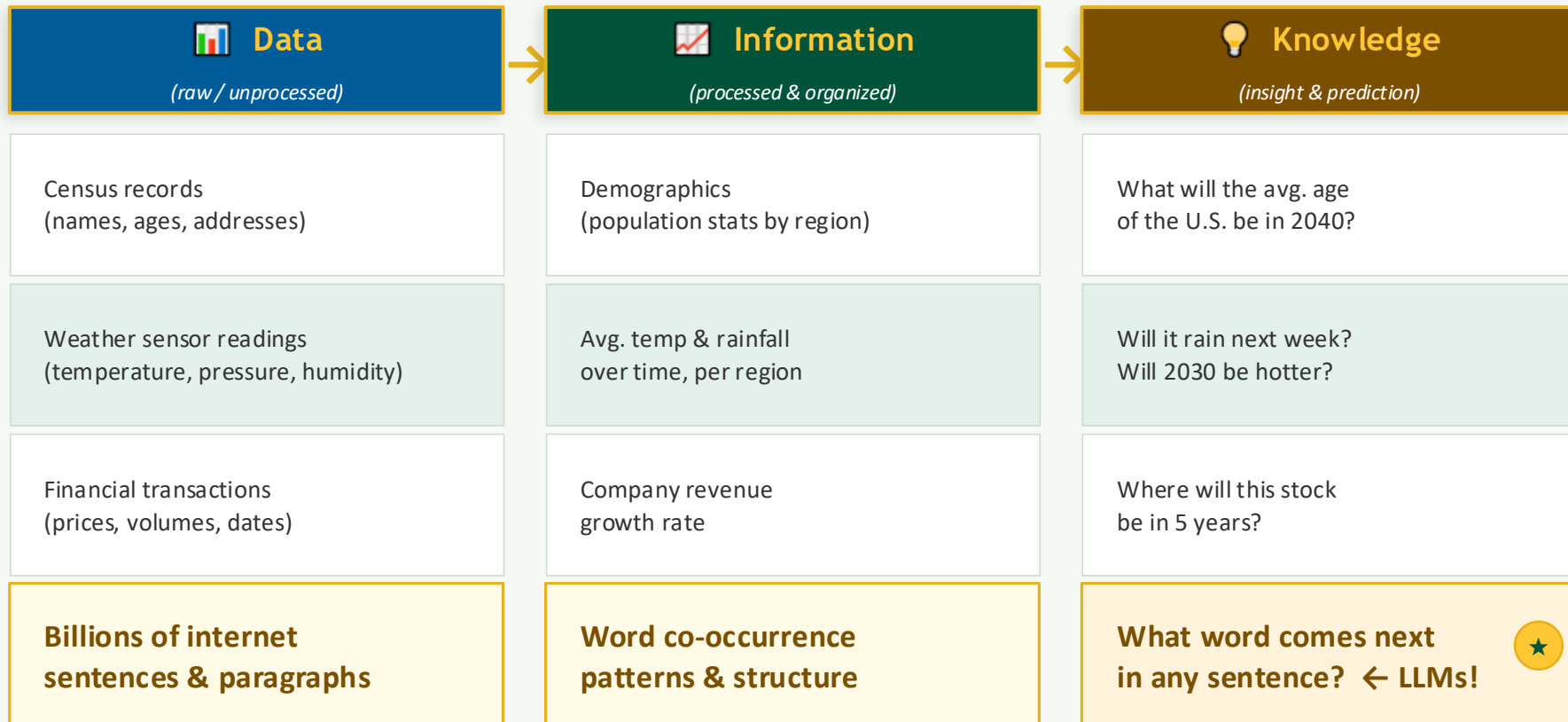
Work individually • There's no wrong answer — just think it through!



Training Data & How Models Learn

How AI went to school — on the entire internet 📖

Data → Information → Knowledge



What Is Training Data?

Training data is the enormous collection of text an AI reads before it can respond to anything. The model studies these examples to learn how language works.

Books



~10 billion words from published books

Web



Hundreds of billions of web pages (CommonCrawl)

Code



Millions of GitHub repositories & coding forums

Research



Millions of academic papers & Wikipedia articles

How Does an AI Model Actually Learn?

During training, the model is shown the beginning of a sentence and asked to predict the next word — billions of times. Each wrong guess causes a tiny adjustment. Over trillions of guesses, the model gets very good.

Example — Fill in the blank:

The capital of France is ____

Paris

She opened the door and ____

walked

The best pizza topping is ____

???



After trillions of fill-in-the-blank attempts, the model becomes very good at predicting text on virtually any topic.

Parameters: The Model's Memory

A parameter (also called a weight) is a number inside the model that gets adjusted during training. LLMs have billions of parameters. These numbers encode everything the model "knows."

GPT-2 (2019)

1.5 Billion

Could write coherent paragraphs

GPT-3 (2020)

175 Billion

Could answer questions, write code

GPT-4 (2023)

~1 Trillion (est.)

Near-human on many standardized tests

Claude 3 (2024)

Undisclosed

Top-tier reasoning and safety focus



More parameters ≠ always better — efficiency, training data quality, and techniques matter just as much.

⚡ OPTIONAL CHALLENGE — Training Data



Pen & Paper

Imagine you are an AI being trained.

You read ONLY these 3 sources:

- Your school textbooks
- Your Instagram feed
- Every recipe ever written

1. What topics would you be GREAT at?
What would you be TERRIBLE at?

2. GPT was trained on the internet.
What problems might that cause?
(Hint: is everything on the internet accurate? Who writes it?)

3. BONUS: Who should decide what data
AI is trained on? Why does it matter?



Try it with an AI Tool

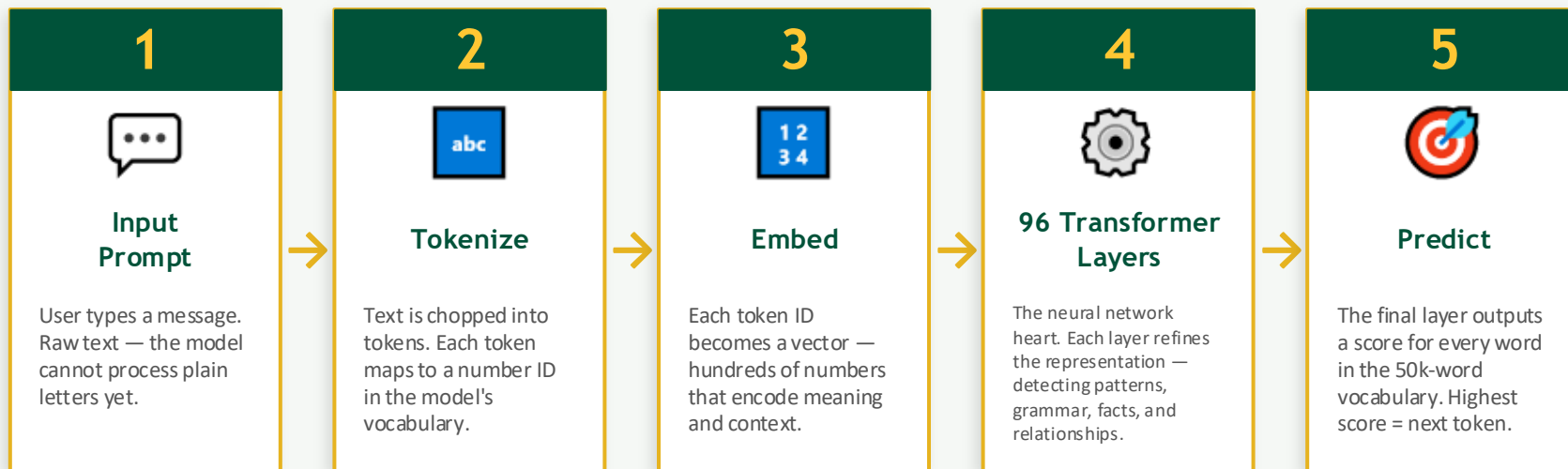
```
# Ask ChatGPT or Claude:
# "What were you trained on?"
#
# Then try:
# "Do you have any biases from
# your training data?"
#
# Finally:
# "Give me an example of something
# your training data got wrong."
```



Predictions & Probability

How AI picks the next word — every single time 🎲

From Tokens to Prediction: The Full Pipeline



► What the data looks like at each step:

"The cat
sat on
the ____"

1284
702
4492
319
278

[0.4, -0.2
1.1, 0.8
-0.5, 1.3
...]

Layer 1
Layer 2
...
Layer 96

mat 72%
floor 15%
hat 13%
→ mat ✓

<https://ig.ft.com/generative-ai/>

AI Is a Next-Word Prediction Machine

Every word ChatGPT generates is the answer to one question: "Given everything typed before this, what word is most likely to come next?"

Prompt: "The sun rises in the ____"



The AI picks "east" — but it could pick any word. It samples from the probability distribution.

Temperature: Creativity vs. Predictability



Low Temperature (e.g., 0.1)

AI always picks the most likely word.

Response is predictable, safe, and consistent.

Best for: Factual questions, math, coding, and anything where accuracy matters.

Q: What is $2 + 2$?

A: 4



High Temperature (e.g., 1.5)

AI picks from a wider range of possible words.

Response is more creative and surprising — sometimes weird.

Best for: Poetry, stories, brainstorming, creative writing.

Q: What is $2 + 2$?

A: A perfect square – the echo of duality itself!

Why AI "Hallucinates"

Hallucination: When AI confidently produces text that sounds correct but is factually wrong or completely made up. Because AI predicts likely words — not true statements — it can't tell when it's lying.

Fabrication

Q: "What books did [obscure author] write?"

A: Invents plausible-sounding book titles with real-looking ISBNs

Fake Citation

Q: "Cite a source for this claim."

A: Generates fake citations with real-looking journal names and page numbers

Outdated Info

Q: "What happened in the news yesterday?"

A: Invents recent events based on past patterns in training data



Always verify AI-generated facts — especially citations, statistics, and anything recent.

⚡ OPTIONAL CHALLENGE — Predictions & Probability



Pen & Paper

Become the AI!

1. Complete this sentence:

"Every morning I wake up and..."

Write 5 different possible next words.

Assign each a probability (%).

They must add up to 100%!

2. Is there ONE right answer?

Why or why not?

3. How is YOUR prediction different from what an AI would predict?

Hint: an AI learned from millions of people — not just you.



Try it with an AI Tool

```
# Go to ChatGPT or Claude.  
# Ask it to SHOW its reasoning:  
#  
# "Complete this sentence and give me  
# the top 5 possible next words with  
# rough probabilities:  
# 'Every morning I wake up and...'"  
#  
# Then try:  
# "Answer as a creative writer"  
# vs.  
# "Answer factually and plainly"  
#  
# Do you notice the difference?
```

Work individually • There's no wrong answer — just think it through!



Prompting

You're the pilot — how you ask changes everything 

What Is a Prompt?

A prompt is the text you give to an AI to get a response. The quality of the prompt directly controls the quality of the answer.

Vague Prompt

`Tell me about climate change.`

Result: A generic 3-paragraph essay about global temperatures and greenhouse gases. Not specific, not useful for any particular purpose.

Specific Prompt

`Explain how rising ocean temperatures affect coral reefs to a 9th grader. Use one real-world example. Keep it under 100 words.`

Result: A focused, grade-appropriate answer with a specific example. Exactly what was needed.

Prompting Techniques That Work



Give It a Role

Telling the AI who to be shapes its tone and vocabulary.

```
"You are a patient math teacher.  
Explain solving for x step-by-step."
```



Specify Format

Control structure: bullet list, table, numbered steps.

```
"List 5 key facts. Keep each  
under 10 words. Use bullet points."
```



Give Context

The more context, the more tailored the answer.

```
"I am a 10th grader studying  
for a chemistry test on acids."
```



Iterate

AI is a conversation partner — keep refining your request.

```
"Make that simpler."  
"Add one more example."  
"Rewrite as a poem."
```

The Hidden Layer: System Prompts

A system prompt is a hidden instruction given to the AI before the user types anything. It defines the AI's role, tone, language, and what it's not allowed to do.

Example — A company's hidden system prompt:

```
# [SYSTEM — hidden from users]
You are a helpful customer support agent
for TechStore. Answer only questions about
our products. Do not discuss competitors.
Always be polite and professional.
If asked about pricing, direct users to:
techstore.com/pricing
```

 This is why ChatGPT, Claude, Siri, and a company chatbot all "feel" different — same technology, different system prompts.

⚡ OPTIONAL CHALLENGE — Prompting



Pen & Paper

The Same Question, 3 Ways:

Task: Get AI to explain photosynthesis.

Write 3 prompts at different levels:

1. WEAK prompt:
(Just type whatever first comes to mind)
2. MEDIUM prompt:
(Add a target audience and a format)
3. STRONG prompt:
(Add role, context, format, and length)

Which one do you predict gives the best answer? Why?



Try it with an AI Tool

```
# Test all 3 prompts in ChatGPT or Claude.  
# Compare the answers side by side.  
#  
# Rate each response 1-10 for:  
#   - Clarity (easy to understand?)  
#   - Accuracy (seems correct?)  
#   - Usefulness (helpful to YOU?)  
#  
# Was your prediction right?  
# What surprised you?
```



AI Ethics & Academia

Using AI responsibly — in school and in life 🏛️

Generative AI Limitations



Hallucinations

AI can generate content that sounds confident but is factually wrong — invented names, fake citations, outdated facts. It doesn't know what it doesn't know.



No True Understanding

Responses are built on statistical patterns, not comprehension. AI cannot reason about cause and effect the way humans do — it predicts plausible text, not correct answers.



Reflects Training Bias

If the training data over-represents certain groups, viewpoints, or languages, the AI output will reflect those skews. Garbage in, garbage out — at billion-parameter scale.



Over-generates / Verbose

AI tends to be wordy and padded. It fills space with confident-sounding language even when a short, honest answer ('I don't know') would be better.



Inconsistent Responses

The same prompt can produce different responses in different sessions. Because outputs are probabilistic, AI is not a reliable calculator — it's a distribution.

The Big Question: Is Using AI Cheating?

The honest answer: it depends on how you use it. AI is a tool — like a calculator or a dictionary. Whether it's cheating depends on the rules, the intent, and what you actually learn.

Academic Dishonesty

- Having AI write your entire essay and submitting it as your own work
- Using AI to answer take-home test or exam questions
- Copying AI output without understanding, attributing, or disclosing it

Productive AI Use

- Using AI to brainstorm ideas, then writing the essay in your own words
- Asking AI to explain a concept you're stuck on — like a tutor
- Using AI to check grammar in writing that you yourself created

Academic Integrity & AI: Know the Rules

Own Your Understanding



If you submit work you don't understand, you're not learning. Employers and colleges will test you — not your AI.

Cite AI Use



If you used AI as a tool, say so. 'I used ChatGPT to brainstorm, then wrote this myself.' Transparency matters.

Use AI as a Tutor



The best academic use: ask AI to explain concepts, quiz you, or give feedback — not to do your work for you.

Consequences Are Real



Schools increasingly use AI detection tools. Being caught can mean a zero, a failing grade, or worse.

AI's Impact on the World



Jobs & Work



Automates repetitive tasks, freeing people for creative and strategic work



Many jobs may be partially or fully replaced — writing, data entry, customer service



Healthcare



AI can detect some cancers earlier than doctors in controlled studies



Privacy risks with medical data; AI errors can have life-or-death consequences



Misinformation



AI can help fact-check content and detect patterns of fake news at scale



AI can also generate convincing fake text, images, and video (deepfakes)



Access & Equity



Free AI tools give everyone — regardless of wealth — access to expert-level help



Those without internet or digital literacy are left even further behind



Which of these matters most to you personally? We'll discuss.

What AI Still Can't Do



Feel Emotions

AI generates text about emotions — it does not experience them.



Have Moral Judgment

AI has no sense of right and wrong — only patterns from training data.



Know Current Events

AI has a training cutoff date — it doesn't know what happened last week.



Learn from Your Chat

Most AI doesn't actually improve from your individual conversations.



Be Truly Creative

AI recombines what it has seen — it doesn't imagine genuinely new ideas.



Guarantee Accuracy

AI is confident even when wrong. Always verify important claims.

⚡ OPTIONAL CHALLENGE – AI Ethics



Pen & Paper

Debate: AI in the Classroom

Scenario: Essay on the American Revolution.

Group A: "Students SHOULD use AI to help write the essay."

Group B: "AI should be COMPLETELY BANNED for this assignment."

Your task:

1. Pick a side
2. Write 3 arguments FOR your position
3. Write 1 counter-argument you must answer

Be ready to share with the class!



Try it with an AI Tool

```
# Use AI to research BOTH sides:
#
# Ask ChatGPT or Claude:
# "Give me 3 strong arguments FOR
# allowing AI in student essays."
#
# Then ask:
# "Now give me 3 strong arguments
# AGAINST allowing AI in essays."
#
# Discuss: Is it ironic to use AI
# to debate whether to use AI?
```

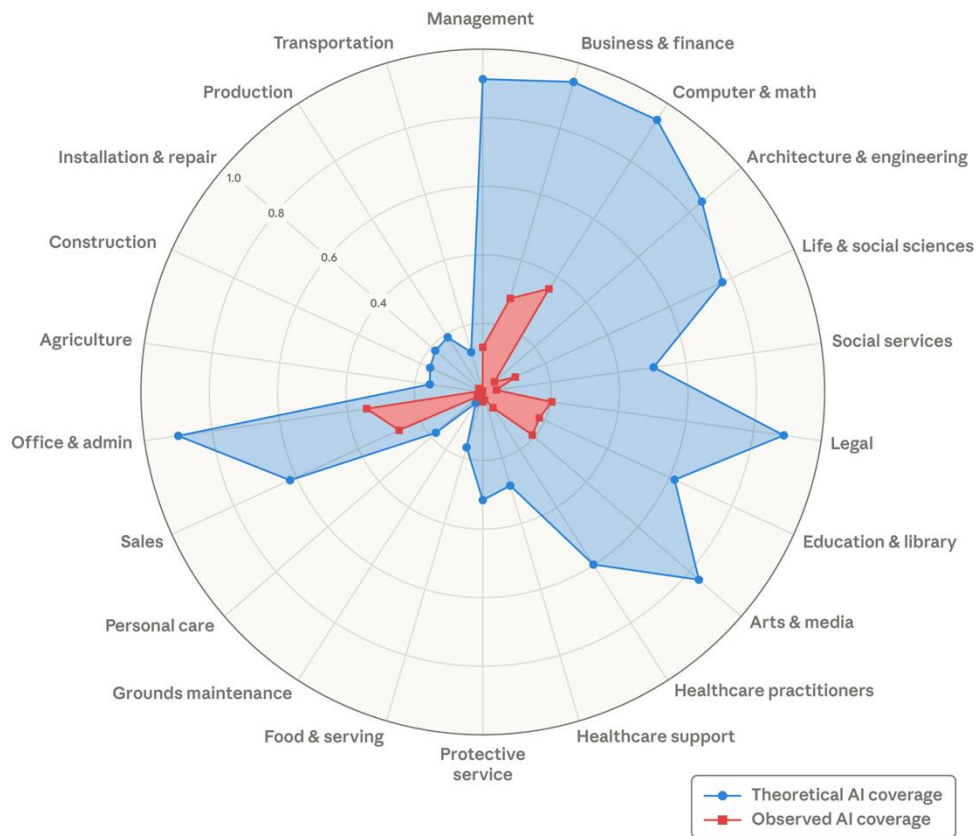
Work individually • There's no wrong answer — just think it through!



Labor Market Impact

The potential impact of artificial intelligence on the labor market 

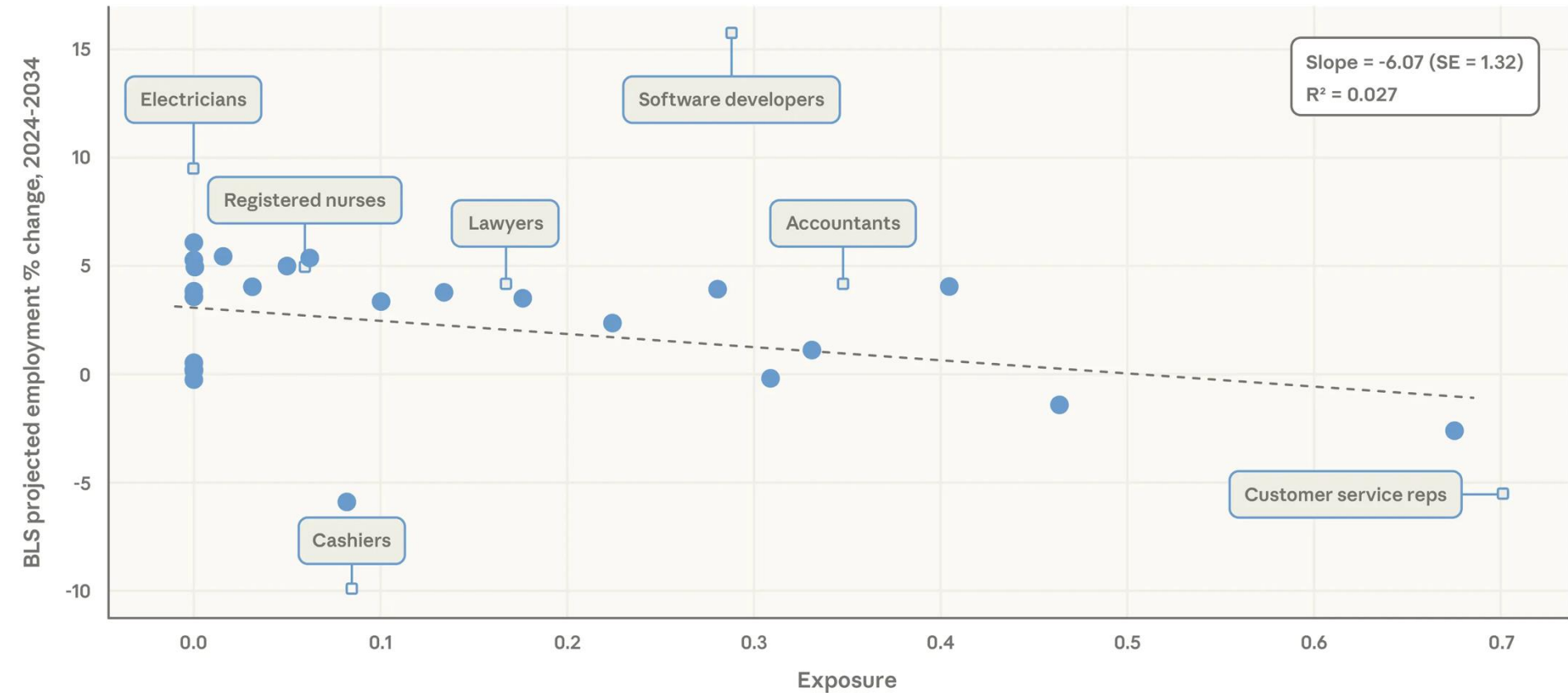
Theoretical Capability and Observed Usage by Occupational Category (Anthropic)



Most Exposed Occupations (Anthropic)

Occupation	Observed exposure	Leading automated task
Computer programmers	74.5%	Write, update, and maintain software programs
Customer service representatives	70.1%	Confer with customers to provide info, take orders, handle complaints
Data entry keyers	67.1%	Read source documents and enter data into systems
Medical record specialists	66.7%	Compile, abstract, and code patient data
Market research analysts and marketing specialists	64.8%	Prepare reports of findings, illustrating data graphically and translating complex findings into written text
Sales representatives, wholesale and manufacturing, except technical and scientific products	62.8%	Contact customers to demonstrate products and solicit orders
Financial and investment analysts	57.2%	Inform investment decisions by analyzing financial information to forecast business, industry, or economic conditions
Software quality assurance analysts and testers	51.9%	Modify software to correct errors or improve performance
Information security analysts	48.6%	Perform risk assessments and test data processing security
Computer user support specialists	46.8%	Answer user inquiries regarding computer software or hardware operation to resolve problems

Projected Employment Growth 2024-2034 (Anthropic)



Observations

- **AI is changing the job market - but not all at once**
 - AI is far from reaching its theoretical capability - actual coverage of job tasks remains a fraction of what's technically feasible
- **Some careers face more risk than others**
 - Computer programmers, customer service representatives, and data entry keyers are among the most exposed occupations, because their tasks can be performed or sped up by AI tools
 - Jobs involving physical work - like cooks, motorcycle mechanics, lifeguards, and bartenders - currently have zero AI coverage
- **Higher education is not a guaranteed shield**
 - People with graduate degrees are nearly four times more represented in the most AI-exposed group compared to the least exposed group
- **Younger workers are feeling early effects**
 - Recent graduates entering heavily AI-affected fields may find it harder to land that first job

Take Away

- The careers you're training for today will exist in a world where AI handles more and more routine tasks. The research suggests that your best strategy is to think carefully about which skills you develop - leaning into **creativity, judgment, physical skill, interpersonal work, and adaptability** - rather than assuming that any single credential or career path is future-proof.

You're Now AI Literate! 🎉



✓ Unlocked

LLM Basics



✓ Unlocked

Tokenization



✓ Unlocked

Training Data



✓ Unlocked

Predictions



✓ Unlocked

Prompting



✓ Unlocked

AI Ethics

Innovation Computer Issues

Front Screen